

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

SSC PART-I (CLASS 9) PRACTICE EXAMINATION

PHYSICS PAPER 17 | ANSWER KEY (Version A)

Syllabus: Unit 7 Density and Temperature, Parts 7A and 7B, that is sections 7.1, 7.2, 7.2.1, 7.2.2, 7.3 and 7.4 (27 marks) + Revision: Unit 6 Work and Energy, including 6.5 Efficiency, and Unit 3 Dynamics-I (11 marks)

Total Marks: 38

Unit 7 Parts 7A and 7B: 27

Revision: 11

Marking rules used on this paper. A missing unit or a missing name loses $\frac{1}{2}$ mark, the same in every question; it never cancels correct working. A correct method with one arithmetic slip keeps the method marks. Where the question says **show that, state, name** or **hence**, the written sentence carries its own marks, as printed below. Page numbers are from **her own printing**, which runs two pages ahead of the online copy.

Where the Aasan Notes quizzes sit on this paper. All six quiz questions from her Part 7A and Part 7B notes are on this paper, with the same numbers she practised: **7A Q1** is Question 2(i) and 2(ii); **7A Q2** is Question 3 (the whole question, 120 g, 50 mL, 160 g); **7A Q3** is Question 2(iii); **7B Q1** is Question 5(A)(iii); **7B Q2** is Question 5(A)(i) and (ii) (175 g, 30 mL to 55 mL); **7B Q3** is Question 4(ii). Those six items are worth **13 of the 27 new marks**, so a student who actually did the quizzes should bank them outright.

SECTION A: ANSWER KEY ($10 \times 1 = 10$ MARKS)

1. 2700 kg/m³ written in g/cm³: (c) 2.7 g/cm³ 1

The book prints " $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$ ", so a density in kg/m^3 is turned into g/cm^3 by dividing by 1000: $2700 \div 1000 = 2.7 \text{ g/cm}^3$. The other three options come from dividing or multiplying by the wrong power of ten, which is the usual slip on this conversion.

Source: Unit 7, section 7.1, For Your Information box, her book p168. SLO P-09-C-01 (define and calculate density).

2. Density of a cube of side 3 cm and mass 81 g: (a) 3 g/cm³ 1

Volume of a cube = (side length)³ = $(3 \text{ cm})^3 = 27 \text{ cm}^3$. $\rho = m / V = 81 \div 27 = 3 \text{ g/cm}^3$. Option (b) 27 g/cm^3 is the volume written as a density, (d) 9 g/cm^3 comes from using side², and (c) 0.33 g/cm^3 is volume divided by mass.

Source: Unit 7, Example 7.1 Density of Regular Shape, her book p169, and the cube volume formula in 7.2.1, her book p171. SLOs P-09-C-01 and P-09-C-02.

3. Density is: (b) a scalar quantity 1

The book closes its definition of density with the plain sentence "**It is a scalar quantity**", because density has a size but no direction. It is a **derived** quantity, not a base one, and it certainly has a unit (kg/m^3 or g/cm^3), so (c) and (d) are both wrong.

Source: Unit 7, section 7.1, the paragraph under Equation 7.1, her book p168. SLO P-09-C-01 (define and calculate density).

4. The volume of a gas: (d) equal to the volume of the container 1

The book's gas point 6: "Gas has no fix shape and volume. Gas spread out in the container and fill it, therefore **its volume is equal to the volume of the container**." A gas has no volume of its own, so options (a), (b) and (c) all contradict the book.

Source: Unit 7, section 7.3 GASES, point 6, her book p175. SLO P-09-C-03 (describe, qualitatively, the particle structure of solids, liquids and gases).

5. Liquids and gases together are called: (c) fluids 1

Point to Ponder: "Because liquids and gases do not maintain a fixed shape, they both have the ability to flow. Thus, they are **collectively referred to as fluids**." The book's own exercise MCQ 5 asks the same thing ("Gases and liquids are categorized as") and its answer is also fluids.

Source: Unit 7, Point to Ponder box, her book p176, and Exercise MCQ 5, her book p186. SLO P-09-C-03.

6. Plasma conducts electric current very well because it contains: (b) free electrons and moving ions 1

The book: "It can highly conduct current because it has **free electrons and moving ions**." A plasma also contains neutral atoms, but it is the free electrons and the moving ions that carry the current.

Source: Unit 7, section 7.4 Plasma as a Fourth State of Matter, her book p176. SLO P-09-C-04 (describe plasma as a fourth state of matter).

7. When electrons are removed from the atoms of a heated gas, the atoms become: (a) positive ions 1

The book: "electrons of the atoms are removed and atoms become **positive ion**. This ionic state of matter is called plasma." An atom that loses a negative electron is left with a positive charge.

Source: Unit 7, section 7.4, her book p176. SLO P-09-C-04.

8. [REVISION, Unit 6] Mass halved and speed doubled: (d) $2 E_k$ 1

$E_k = \frac{1}{2}mv^2$. New $E_k = \frac{1}{2} \times (m/2) \times (2v)^2 = \frac{1}{2} \times (m/2) \times 4v^2 = 2 \times (\frac{1}{2}mv^2) = 2 E_k$. Doubling the speed alone multiplies E_k by 4, and halving the mass alone halves it, so the two together give $4 \times \frac{1}{2} = 2$. Option (c) $4 E_k$ is the trap: it ignores the halved mass.

Source: Unit 6, section 6.2.1 Kinetic Energy, book p142 to p144, where $E_k = \frac{1}{2}mv^2$ is derived, and Exercise Short Response Question 4, book p165. SLOs P-09-B-64 and P-09-B-66. This re-tests her OPF First Term Physics MCQ 9, which used the same half mass and double speed.

9. [REVISION, Unit 6] Water held in a dam reservoir stores: (c) gravitational potential energy 1

The book: "Hydroelectric generation is the process of using the **gravitational potential energy** of water to generate electric energy. The force of gravity pulls the water down, giving it a lot of kinetic energy. This kinetic energy is then converted into electric energy by large turbines." Figure 6.14 is labelled Gravitational Potential Energy. Electrical energy is the **output**, not what the stored water holds.

Source: Unit 6, section 6.3.4 and Figure 6.14, book p155; energy names from Table 6.1, book p142. SLO P-09-B-71 (describe how useful energy may be obtained from natural resources). This re-tests her OPF First Term Physics MCQ 11, where she chose electrical energy.

10. [REVISION, Unit 3] Mass of a stone that weighs 540 N, with $g = 10 \text{ N/kg}$: (b) 54 kg 1

The book prints $W = mg$, or $m = W / g$. So $m = 540 \div 10 = 54 \text{ kg}$. Weight is a force in newtons and mass is in kilograms; the weight value itself (540) is never the mass.

Source: Unit 3, section 3.4 Gravitational Field and Example 3.2, book p78 and p79. SLOs P-09-B-19 and P-09-B-20. This re-tests her OPF First Term Physics 2(ii), where a weight of 600 N was used as if it were a mass.

Answer letters: 1(c), 2(a), 3(b), 4(d), 5(c), 6(b), 7(a), 8(d), 9(c), 10(b). Balance: a = 2, b = 3, c = 3, d = 2, with no letter repeated in consecutive questions and no run of a repeating pattern.

SECTION B: ANSWER KEY ($3 \times 4 = 12$ MARKS)

2. Density: definition, equation, volume formulas, and why size does not change density. 4

Aasan Notes Part 7A, Quiz 1 is parts (i) and (ii); Quiz 3 is part (iii).

(i) Book definition: "Mass per unit volume of the substance is called its density." $\frac{1}{2}$

Equation 7.1: $\rho = m / V$, where ρ (rho) is the density, m is the mass of the substance and V is its volume. $\frac{1}{2}$

SI unit: kg/m^3 (kilogram per cubic metre). $\frac{1}{2}$ g/cm^3 alone, without the SI unit, loses this $\frac{1}{2}$.

(ii) Volume of a cuboid = Length $\times$ Width $\times$ Height $\frac{1}{2}$

Volume of a cube = **(side length) 3** $\frac{1}{2}$

Volume of a sphere = **$(4/3) \pi r^3$** $\frac{1}{2}$ r is the radius, that is half the diameter. Writing the sphere formula with d instead of r loses this $\frac{1}{2}$.

(iii) Density is mass per unit volume, so it belongs to the **material** and not to how much of it there is. The large cube does have more mass, but it also has proportionally more volume, so the ratio m / V comes out the same. 1

The book's own sentence, which earns full credit on its own: "Density of small amount of a substance is the same as the density of its bulk because density is calculated by the amount of matter in its unit volume."

Source: Unit 7, section 7.1, her book p168 (definition, Equation 7.1, SI unit, bulk sentence); section 7.2.1 volume formulas and Equation 7.3, her book p171. SLOs P-09-C-01 and P-09-C-02.

3. Density of a liquid: empty cylinder 120 g, 50 mL of liquid, total 160 g. 4

Aasan Notes Part 7A, Quiz 2, with exactly the numbers she practised.

(i) Mass of liquid = Mass of measuring cylinder and liquid – Mass of empty cylinder ½

Mass of liquid = 160 g – 120 g = 40 g ½

(ii) Equation 7.2: Density of liquid = Mass of liquid / Volume of liquid ½

$\rho = 40\text{ g} \div 50\text{ mL} = 0.8\text{ g/mL}$ 1 (unit required; ½ off if missing)

(iii) 1 mL = 1 cm³, so 0.8 g/mL = 0.8 g/cm³. To convert g/cm³ into kg/m³, multiply by 1000. ½

$0.8 \times 1000 = 800\text{ kg/m}^3$ ½

The closing sentence, which carries its own ½: "The density of the liquid is 800 kg/m³, that is 0.8 g/mL." ½

Marking notes. The subtraction in (i) is the step the whole question turns on: a student who divides 160 by 50 and writes 3.2 g/mL loses (i) and (ii) but can still earn (iii) by converting her own value correctly. This question never asks her to name the liquid; if she adds on her own that Table 7.1 gives 800 kg/m³ for petrol, accept it and neither add nor cut a mark, because the book's two pages disagree about petrol. Source: Unit 7, section 7.2 Density of Liquids, the five steps, the mass-of-liquid line and Equation 7.2, her book p170; Example 7.2, her book p171; the conversion rule, her book p168. SLOs P-09-C-01 and P-09-C-02.

4. The three states of matter. 4

Aasan Notes Part 7B, Quiz 3 is part (ii).

(i) One mark for each complete row (½ for each correct cell):

	Solid	Liquid	Gas
(a) Density	High	Medium	Low
(b) Arrangement of particles	Regular pattern	Randomly arranged	Randomly arranged
(c) Movement of particles	Vibrate around a fixed position	Move around each other	Move quickly in all directions

3 Table 7.2's own wording is printed above. Equivalent wording from section 7.3.1 is fully accepted: "fixed lattice pattern" or "closely packed" for a solid's arrangement, "flow and glide over each other" for a liquid's movement, "constant random motion, colliding with each other and with the walls" for a gas, and "intermediate" for a liquid's density.

(ii) In a **solid** the attractive force between the particles is **strong**, so the particles are closely packed with very small gaps; the same volume therefore holds much more matter and the density is high. ½

In a **gas** the attractive force is **negligibly weak**, so the distance between the particles is even larger than the size of the particles themselves; the same volume holds far less matter, so the density is low. ½

Source: Unit 7, Table 7.2 States of Matter, her book p175; section 7.3 opening ("The difference in strength of this force is the reason for smaller or large gaps between particles"), her book p174; section 7.3.1 SOLIDS points 2, 3 and 6 and GASES points 2, 3 and 7, her book p174 and p175; Figure 7.1 caption ("Solids have more atoms per unit volume than liquids and gases"), her book p168. SLO P-09-C-03. Do not set a question off Figure 7.4: in this printing its "Solids" and "Gases" panels are swapped.

SECTION C: ANSWER KEY (2 × 8 = 16 MARKS)

5. Finding the density of solids. 8

Aasan Notes Part 7B, Quiz 2 is (A)(i) and (ii); Quiz 1 is (A)(iii).

(A) The stone, mass 175 g, water 30 mL to 55 mL

(i) Volume of object = $V_f - V_i = 55 \text{ mL} - 30 \text{ mL} = 25 \text{ mL} = 25 \text{ cm}^3$ 1 (1 mL = 1 cm³)

(ii) Formula: $\rho = \text{Mass of solid} / \text{Volume of solid}$ ½

$\rho = 175 \text{ g} \div 25 \text{ cm}^3 = 7 \text{ g/cm}^3$ 1 (unit required; ½ off if missing. 7.0 g/cm³ is the same answer.)

(iii) The method is the **displacement method**. ½

It is used for an **irregular shaped solid object**, such as a stone. ½

The two conditions the book gives: the object must **sink in water**, and it must be **insoluble** (it must not dissolve in the water). ½ (½ each; one condition alone earns ¼)

(B) The rectangular metal block, 20 cm × 10 cm × 5 cm, mass 19.3 kg

(iv) length = 20 cm = **0.20 m**, width = 10 cm = **0.10 m**, height = 5 cm = **0.05 m** ½

Volume of cuboid = Length × Width × Height = $0.20 \times 0.10 \times 0.05 = 0.001 \text{ m}^3$ 1

(v) $\rho = m / V = 19.3 \text{ kg} \div 0.001 \text{ m}^3 = 19300 \text{ kg/m}^3$ 1 (unit required; ½ off if missing)

(vi) Table 7.1 gives gold as 19300 kg/m³, so the metal is **gold**. 1

The closing sentence, which carries its own ½: "The density of the block is 19300 kg/m³, so the metal is gold." ½

Marking notes. In (iv) to (vi) the marks follow the working: a wrong volume used correctly in the next step still earns the later marks. Working in g and cm³ throughout is fully correct and keeps every mark: $V = 20 \times 10 \times 5 = 1000 \text{ cm}^3$ and $\rho = 19300 \text{ g} \div 1000 \text{ cm}^3 = 19.3 \text{ g/cm}^3$, which is 19300 kg/m³ and still gold, provided the unit is written. Source: Unit 7, section 7.2.2 Density of Irregular Shaped Object, the six steps, "stones which can sink in the water", "any insoluble irregular shaped object" and Equation 7.4, her book p172; Figure 7.3 and Example 7.4, her book p173; section 7.2.1 volume formulas and Equation 7.3, her book p171; Example 7.3 Density of Brick, which converts cm to m in the same way, her book p172; Table 7.1, her book p169. SLOs P-09-C-01 and P-09-C-02.

6. [REVISION, Unit 6, section 6.5] Efficiency. 8

(i) Book definition: **efficiency is the ratio of the useful work output to the total work input** (in terms of energy, the ratio of the useful energy output to the energy input). 1

$\eta = (E_{\text{out}} / E_{\text{in}}) \times 100\%$ (efficiency has no unit, because it is a ratio of the same quantities) 1

(ii) $\eta = E_{\text{out}} / E_{\text{in}} = 1500 \text{ J} \div 2500 \text{ J} = 0.6$ 1

% efficiency = $0.6 \times 100\% = 60\%$ 1

(iii) $P_{\text{out}} = \eta \times P_{\text{in}} = 0.6 \times 600 \text{ W}$ 1

$P_{\text{out}} = 360 \text{ W}$ 1 (unit required; ½ off if missing. Using 60 instead of 0.6 gives 36,000 W and loses this mark.)

(iv) A machine cannot be 100% efficient because **friction always lowers its efficiency**: some of the input energy is always wasted, usually as heat, so the work output is always less than the work input. 1

The device with the lowest efficiency in Table 6.3 is the **incandescent light**, at 5%. 1

Source: Unit 6, section 6.5 Efficiency, equations 6.7 to 6.10 and Figure 6.21, book p160 and p161; Table 6.3 Typical Efficiencies of Energy Transformation Technologies, book p162, checked cell by cell on her own scan; Example 6.6 uses the same method on a petrol engine. SLOs P-09-B-74, P-09-B-75 and P-09-B-76.

Mark distribution check

Section	Unit 7, Parts 7A and 7B (new)	Revision	Total
A: MCQs 10 × 1	7 (Q1 to Q7)	3 (Q8, Q9: Unit 6; Q10: Unit 3)	10
B: Short 3 × 4	12 (Q2, Q3, Q4)	0	12
C: Long 2 × 8	8 (Q5)	8 (Q6: Unit 6, section 6.5)	16
Total	27 (71%)	11 (29%)	38

Per question: new = MCQs 1 to 7, Q2, Q3, Q4, Q5. Revision = MCQs 8, 9, 10 and Q6.

Section coverage: 7.1 in MCQs 1 and 3 and in Q2(i) and Q2(iii); 7.2 Density of Liquids in Q3 (all 4 marks); 7.2.1 in MCQ 2, Q2(ii) and Q5(B); 7.2.2 in Q5(A); 7.3 in MCQs 4 and 5 and in Q4; 7.4 in MCQs 6 and 7.